My name is Minenhle Dlamini, and I am an independent civic technology builder in Pietermaritzburg, South Africa. I design & develop digital systems that aim to solve structured problems in public systems such as education access and civic participation.

Currently, I am building two prototype systems:

i. *A Student Pathway Recommendation Platform designed to help Grade 12 learners make better post-school decisions.*
ii. *A secure Online Voting System designed to improve accessibility and transparency in elections.*

The Student Pathway Platform is the focus of this document because it is the most mature and ready for pilot testing. My goal is to build technology that reduces confusion, improves decision-making, and increases opportunity access for young people entering higher education.

Problem Statement

Every year, thousands of Grade 12 learners in South Africa struggle to decide what they should pursue after finishing school.

Many Students:

i. *Do not understand which courses match their academic strengths.*
ii. *Apply to programs they do not qualify for.*
iii. *Miss application deadlines.*
iv. *Choose paths based on pressure rather than informed decisions.*

This results in several challenges:

i. *High numbers of misdirected applications.*
ii. *Students being rejected from institutions due to poor alignment.*
iii. *Increased gap years caused by misapplications.*
iv. *Learners feeling lost or discouraged about their future.*

While guidance exists through teachers and counsellors, many schools lack scalable digital tools that help students analyse their academic performance and explore realistic study options. This creates a gap where students must make life impacting decisions without structured guidance.

Proposed Solution

The Student Pathway Recommendation Platform is a digital tool designed to assist Grade 12 learners in identifying realistic academic pathways based on their academic results.

The platform allows students to:

i. *Create a basic student profile*
ii. *Enter their academic results*
iii. *Receive course recommendations aligned with their strengths*
iv. *View institutions that offer those courses*
v. *Explore potential academic pathways before submitting applications.*

The system analyses student performance and presents structured recommendations that help learners:

i. *Understand their strengths*
ii. *Explore relevant study options*
iii. *Make more informed application decisions*

The platform does not replace teachers or career counsellors but rather acts as a decision-support tool that enhances guidance resources already available in schools.

Technology Overview

The System is currently built as a full stack web application with the potential to expand into a mobile platform.

Key technical components include:

1. *Backend*
    - *Node.js server*
    - *Express framework*
    - *Database hosted through MongoDB.*

2. *Front-end*
    - *React-based web interface.*

3. *Hosting*
    - *Cloud hosting*
4. *Security considerations include:*
    - *Secure database connections*
    - *Authentication systems for administrative access*
    - *Data protection measure to safeguard student information*
5. *The Platform has already implemented core functionality including:*
    - *Student Profile Creation*
    - *Data entry for academic results*
    - *Course Recommendation Logic*
    - *Institution Suggestion features*
    - *Administrative dashboard for monitoring student data*

The System is currently being refined and prepared for pilot testing.

Pilot Rollout Plan

The first phase of the platform will focus on pilot testing within a small number of high schools.

Pilot objectives include

- *Testing usability among Grade 12 learners*
- *Measuring student engagement with the platform*
- *Collecting feedback from teachers & students*
- *Evaluating whether the platform improves decision-making confidence.*

The Initial pilot plan includes

i. *Introducing the system at a participating high school.*
ii. *Demonstrating how students can use the platform*
iii. *Allowing Grade 12 learners to test the system*
iv. *Collecting feedback through surveys*

Success metrics for the pilot include:

- *At least 30% of Grade 12 learners using the system*
- *At least 25% of users reporting increased confidence in their course choices*
- *Evidence that some learners adjusted their study plans after receiving recommendations.*

These metrics will help evaluate whether the platform provides real value to students.

Long Term Vision

If Pilot testing proves successful, the platform could expand in several ways.

Expansion opportunities include:

- *Integration with additional schools*
- *Partnerships with higher education institutions*
- *Improved recommendation algorithms*
- *Mobile application availability for wider access*

Over time the system could become a national digital guidance tool that helps students make informed decisions before applying to universities or colleges.

The broader vision is to contribute to an education ecosystem where students are better informed, applications are more aligned with student capabilities and fewer learners lose opportunities due to confusion on lack of guidance.